DE–2315

Sub. Code
13

DISTANCE EDUCATION

B.Sc. (Computer science) DEGREE EXAMINATION, DECEMBER 2015.

DIGITAL COMPUTER FUNDAMENTALS

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the 1's and 2's complement with example. (10)

   (b) Convert the following:

   (i) $(ab5.cd)_{16} = ()_{10}$

   (ii) $(101110.0101)_2 = ()_{10}$

   (iii) $(456.78)_{10} = ()_{16}$. (10)

2. (a) Simplify the following Boolean function to the minimum number of literals. $(a+b)'(a'+b')'$. (10)

   (b) Simplify using K map

   $F(a,b,c,d) = \Sigma(2,3,12,13,14,15)$. (10)

3. (a) State and prove DeMorgan's theorem. (10)

   (b) Apply DeMorgan's theorem to the following expression

   (i) $[(A + B)' + C]'$

   (ii) $[(AB + AC)' + (AB)' C]'$. (10)

4. (a) Design a full adder circuit. (10)

   (b) Draw the logic diagram of master-slave flip-flop and explain. (10)

5. (a) What is a multiplexer? Explain a 4 to 1 line multiplexer in detail. (10)

   (b) Explain the working of a BCD adder along with the functional logic diagram. (10)

6. (a) Design a RS flip-flop using NAND gates and explain. (10)

   (b) Explain the shift Register with neat diagram. (10)

7. (a) Discuss the basic operation of a Binary counter. (10)

   (b) Write short notes on Cache memory. (10)

8. (a) Explain the organization of magnetic disk memory. (10)

   (b) Describe the different types of RAM. (10)

———————

DE–2316

Sub. Code
14

DISTANCE EDUCATION

B.Sc. (CS) DEGREE EXAMINATION, DECEMBER 2015.

DATA STRUCTURES USING C

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the various types of constants available in C with example.

   (b) Write the syntax for 'FOR' loop and explain how it works with an example.

2. (a) Write a 'C' program to find the biggest of two numbers using ternary operator.

   (b) Explain SWITCH statement with its general form and give one example.

3. (a) Given an array of N elements, write a 'C' program to find the biggest and smallest numbers in the array.

   (b) What is macro? Write macros to

      (i) Check whether is a lower alphabet

      (ii) Convert the lower case alphabet into upper case alphabet using the macro.

4. (a) Write a C program to generate Fibonacci series upto 25 terms using recursion

(Fibonacci series : 0 1 1 2 3 5 8…)

(b) Explain the role of the C preprocessors.

5. (a) What is structure? Explain how a structure is declared by giving an example.

(b) Write a 'C' program to arrange the given set of 'n' integer numbers in descending order using pointers for the array.

6. (a) Explain the following functions :

(i) getc

(ii) f open

(iii) f close

(iv) f seek

(v) f tell.

(b) What is dynamic memory allocation? How does it help in building complex programs? Explain.

7. (a) Explain stack and various operations performed on it.

(b) Explain how to carry out insertions and deletions in a queue.

8. (a) Explain in detail about doubly linked list with insertion and deletion algorithm.

(b) List the methods of binary tree representation, with a suitable example explain any one.

DE–2317

Sub. Code
15

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, DECEMBER 2015.

DISCRETE MATHEMATICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 =100)

1. (a) Verify whether $Q \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$ is a tautology.

   (b) Show that $(P \to (Q \to R)) \Rightarrow ((P \to Q) \to (P \to R))$.

2. (a) Show that $\neg(P \leftrightarrow Q) \Rightarrow (P \vee Q) \wedge \neg(P \wedge Q)$.

   (b) Prove that $\{\uparrow\}$ is functionally complete.

3. (a) Show that $A \cap (B - C) = (A \cap B) - (A \cap C)$.

   (b) If R and S are relations from A to B show that $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$.

4. (a) Let $A = \{a, b\} \cdot R = \{(a,b), (b,a), (b,b)\}$ and $S = \{(a,a), (b,a), (b,b)\}$ be relations on A. Find $S \circ R$ and $R \circ S$.

   (b) Prove that the relation "congruence modulo m" over the set of positive integers is an equivalence relation.

5. (a) Let $f : A \to B$ and $g : B \to C$ be both one-to-one and onto functions. Then prove that $f^{-1} \circ g^{-1} = (g \circ f)^{-1}$.

(b) Prove by induction

$$1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + ... + n \cdot (n+1) = \frac{n(n+1)\,(n+2)}{3}.$$

6. (a) Show that identity element of a group is unique.

(b) If $(a * b)^2 = a^2 * b^2$ for all $a, b \in G$, then show that G is abelian.

7. (a) State and prove Lagrange's theorem on group.

(b) Show that the number of vertices of odd degree in an undirected group is even.

8. (a) Define regular graph. Can a regular graph be a complete graph?

(b) Prove that the maximum number of edges in a simple disconnected graph G with n vertices and K components is $\frac{(n-K)\,(n-K+1)}{2}$.

DE–2318

Sub. Code
23

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

OBJECT ORIENTED PROGRAMMING AND C++

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Explain briefly about the important features of Object Oriented Programming with neat sketch. (10)

   (b) Compare and contrast the swapping program written in C and C++. (10)

2. (a) What is a class? How does it accomplish data hiding? Discuss briefly. (10)

   (b) Discuss in detail about friend function with appropriate examples. (10)

3. (a) Give a detailed note on Pointers with suitable illustrations. (10)

   (b) What is copy constructor? When it is used? Explain with examples. (10)

4. (a) Explain briefly about polymorphism with appropriate illustrations. (10)

   (b) What is operator overloading? Explain how to overload unary and binary operator. (10)

5. (a) Explain briefly about the various forms of inheritance with neat sketch. (10)

   (b) Describe the importance of virtual base class with suitable illustrations. (10)

6. (a) How Object Oriented Programming is different from procedure oriented programming? Explain. (10)

   (b) Explain in detail about various types of operators with neat sketch. (10)

7. (a) Write a C++ program to illustrate the concept of this pointer. (10)

   (b) Write a brief note on function overloading. (10)

8. (a) Discuss briefly how polymorphism is achieved in C++ with suitable example. (10)

   (b) Write a C++ program to find the second largest element and second smallest element from the given n numbers. (10)

DE–2319

Sub. Code
24

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (C.S) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

SCIENTIFIC COMPUTING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Solve the system of equations $2x + 3y - z = 5$; $4x + 4y - 3z = 3$; $2x - 3y + 2z = 2$ by Gauss–elimination method. (10)

   (b) Solve the system of equations $10x - 5y - 2z = 3$; $4x - 10y + 3z = -3$; $x + 6y + 10z = -3$ by Gauss– Jacobi method. (10)

2. (a) Using Lagrange's interpolation formula find $y(10)$, given that $y(5) = 12$; $y(6) = 13$; $y(9) = 14$; $y(11) = 16$. (10)

   (b) The following data are taken from the steam table:

Temp °C:	140	150	160	170	180
Pressure kg^4 / cm^2:	3.685	4.854	6.302	8.076	10.225

   Find the pressure at temperature $t = 142°$ and $t = 175°$. (10)

3. (a) Evaluate $\int_{4}^{5.2} \log_e x \, dx$ using Trapezoidal rule and Simpson's one–third rule with $h = 0.2$. (10)

(b) Find the first and second derivatives of $f(x)$ at $x = 1.5$ if (10)

x:	1.5	2.0	2.5	3.0	3.5	4.0
y:	3.375	7.0	13.625	24.0	38.875	59.0

4. (a) Find the mean, median and mode of the following frequency distribution: (10)

Class:	20–24	25–29	30–34	35–39	40–44	45–49	50–54	55–59
Frequency:	3	5	10	20	12	6	3	1

(b) Calculate the mean and standard deviation for the data: (10)

x:	25	35	45	55	65	75	85
f:	3	61	132	153	140	51	2

5. (a) Fit a parabola of the type $y = a + bx + cx^2$ to the following data: (10)

x:	10	15	20	25	30	35	40
y:	11	13	16	20	27	34	41

(b) The contents of 3 urns are

Urn I : 1 white, 3 red, 2 black balls

Urn II : 3 white, 1 red, 1 black balls

Urn III : 3 white, 3 red, 3 black balls.

Two balls are chosen from a randomly selected urn. If the balls are 1 white and 1 red ball, what is the probability that they come from Urn III. (10)

6. (a) A random variable X has the following probability functions.

x_i :	–2	–1	0	1	2	3
$p(x_i)$:	0.1	K	0.2	2K	0.3	K

Find (i) The value of K (ii) mean and variance. (10)

(b) (i) Define moment generating function.

(ii) A random variable X has the probability function $p(x) = \frac{1}{2^x}$; $x = 1,2,3.....$, find its moment generating function. (10)

7. (a) A random sample of 10 boys has the following I.Q 70, 120, 110, 101, 88, 83, 95, 98, 107, 100.

Do these data support the assumption of a population mean I.Q of 100? $[\gamma = 9, t_{0.05} = 2.26]$. (10)

(b) In one sample of 8 observations the sum of the squares of deviations of the sample values from the sample mean was 84.4 and in another sample of 10 observations it was 102.6. Test whether the difference in variances is significant at 5% level using F–Test. [$F_{(7,9)}$ at 5% level = 3.29]. (10)

8. (a) Apply χ^2 – test of goodness of fit to the following data

Observed frequency :	1	5	20	28	42	22	15	5	2
Theoretical frequency :	1	6	18	25	40	25	18	6	1

[ψ^2 for 6 d.f at 5% level is 12.59]. (10)

(b) Find the mean deviation about median for the following data: (10)

x:	58	60	62	64	66	59	61	63	65
f:	1	32	33	20	8	20	35	22	10

DE–2320

Sub. Code
25

DISTANCE EDUCATION

B.Sc. (C.S.)/B.Sc. (C.S.) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

APPLICATION PROGRAMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write down the steps needed to create, open and save files in windows. (10)

   (b) Explain briefly about windows desktop and toolbars. (10)

2. (a) Write short notes on templates and wizards. (10)

   (b) What are the features of mail merge in Ms-word? (10)

3. (a) What are the different types of charts in Ms-excel? Explain. (10)

   (b) Describe in detail about building an excel database. (10)

4. (a) How will you create and edit slides in Ms-Powerpoint? (10)

   (b) Write short notes on handouts. (10)

5. (a) What are the steps needed to create reports in Ms-access? (10)

   (b) Discuss in detail about creation of primary keys. (10)

6. (a) Write brief notes on dialog boxes in windows. (10)

   (b) Explain in detail about print preview and page orientation in Ms-word. (10)

7. (a) How will you do formatting the text in Ms-excel? (10)

   (b) Discuss briefly about background and color scheme in Ms-powerpoint. (10)

8. (a) Explain about combine excel data and charts with word documents. (10)

   (b) Describe in detail about formatting chart series in Ms-excel. (10)

———————

DE–2321

Sub. Code
31

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

COMPUTER GRAPHICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss on the applications of computer graphics. (10)

   (b) Write an algorithm for circle generation. (10)

2. (a) How cathode ray tube can be used for display device? (10)

   (b) Describe Brehenham's Line Drawing Algorithm. (10)

3. (a) How a two dimensional object is to be rotated about origin in clockwise direction. Explain. (10)

   (b) Explain the technique for clipping a convex polygon. (10)

4. (a) Give the matrix representation for two dimensional transformations. (10)

   (b) Describe the Sutherland Hodgman Algorithm. (10)

5. (a) Mention the concepts involved in the three dimensional transformations. (10)

(b) How scaling is done for a three dimensional object? (10)

6. (a) How viewing process is done for a three dimensional object? (10)

(b) Discuss on Z-Buffer Algorithm. (10)

7. (a) How to construct a three dimensional view? (10)

(b) Mention the need for depth comparison. (10)

8. (a) Discuss on the components of User Interface. (10)

(b) Give the importance of feedback. (10)

DE–2322

Sub. Code
32

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, DECEMBER 2015.

VISUAL PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss the format of a window with diagram. (10)

   (b) Write about menus and menu operations. (10)

2. (a) Discuss the different ways to run an application from windows. (10)

   (b) Write briefly about the features of control panel. (10)

3. (a) Explain the Message Box and Input Box functions with example. (10)

   (b) Explain briefly about standard toolbar. (10)

4. (a) Explain the select-case statement with example. (10)

   (b) Explain the properties of text box and labels. (10)

5. (a) Write about line and shape controls. (10)

   (b) Explain MDI forms with example. (10)

6. (a) Discuss about selecting and resizing windows. (10)

   (b) Write about the drawing tools available in paintbrush. (10)

7. (a) Explain the following :

   (i) List box

   (ii) Combo box. (10)

   (b) Explain any two loop statements with example. (10)

8. (a) Explain about control arrays with example. (10)

   (b) Explain event-driven programming with a sample program. (10)

———————

DE-2323

Sub. Code
33

DISTANCE EDUCATION

B.Sc. (C.S)/B.Sc. (C.S) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

OPERATING SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What are the system components of an operating system and explain them? (10)

   (b) How protection is provided for the hardware resources by the operating system. (10)

2. (a) Write about the various CPU scheduling algorithms. (10)

   (b) Explain what semaphores are, their usage, implementation given to avoid busy waiting and binary semaphores. (10)

3. (a) Discuss about the message passing technique. (10)

   (b) What is the use of monitors for resource allocation? (10)

4. (a) Give a detailed description about deadlocks and its characterization. (10)

   (b) Discuss the methods of Deadlock detection and Recovery. (10)

5. (a) Write short notes on space I/O software. (10)

   (b) Explain about the methods used to prevent deadlocks. (10)

6. (a) Discuss how memory is allocated in variable partition multiprogramming. (10)

   (b) Discuss segmentation in detail. Compare it with paging. (10)

7. (a) What is meant by free space management? Explain. (10)

   (b) Explain the concept of paging. Use any one method and explain how PMT's implemented. (10)

8. (a) Explain the services provided by a kernel I/O subsystem. (10)

   (b) Explain and compare the FCFS and SJF disk scheduling algorithms. (10)

———————

DE–2324

Sub. Code
34

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (Computer Science) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

INTERNET PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the Network Topologies with an example. (10)

   (b) Explain the Network and Transport layer. (10)

2. (a) Discuss on Mail transfer and File transfer protocol in detail. (l0)

   (b) Explain Finger user information protocol in detail. (10)

3. (a) Describe the various Internet Applications with suitable example. (l0)

   (b) What is Internet programming and how is it different from Intranet programming? (10)

4. (a) How to create web server? Explain with example. (10)

   (b) Explain HTTP methods in detail. (10)

5. (a) Explain various kinds of data types in Java. (10)

   (b) Write a Java program to sort the 'n' numbers in descending order. (10)

6. (a) Explain various types of operators in JAVA with suitable example. (10)

   (b) Discuss on various types of statements in JAVA with example. (10)

7. (a) Write a note on Java classes and objects with example. (10)

   (b) Write a program in JAVA to calculate simple and Compound interest. (10)

8. (a) Explain String handling in JAVA with example. (10)

   (b) Write a JAVA program to read a string and print its length. (10)

DE–2325

Sub. Code
35

DISTANCE EDUCATION

B.Sc.(CS)/ B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

RDBMS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Describe the advantages of Database Management System. (10)

   (b) Explain different user interfaces. (10)

2. (a) Briefly discuss on Relational model. (10)

   (b) Explain Data manipulation facilities in detail. (10)

3. (a) Discuss on various types of databases using oracle. (10)

   (b) Discuss on client/server systems. (10)

4. (a) Explain the following in detail :

      (i) Database objects. (5)

      (ii) Data Access SQL commands. (5)

   (b) How will you join multiple tables in a query? Explain with examples. (10)

5. (a) Explain the following with examples :

   (i) Copying another table. (5)

   (ii) Dropping a table. (5)

   (b) Discuss on Embedded SQL statements. (10)

6. (a) Briefly discuss on Embedded SQL. (10)

   (b) How will you describe table definitions? Discuss with example. (10)

7. (a) How is table created? Explain with examples. (10)

   (b) How is table modified and renamed? Explain with examples. (10)

8. (a) Explain the mapping operations on Relational model. (10)

   (b) Explain Data control facilities in detail. (10)

DE–9031

Sub. Code
16

DISTANCE EDUCATION

B.Sc. DEGREE EXAMINATION, DECEMBER 2015.

Computer Science

COMPUTER LAB I – C AND DATA STRUCTURES

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the students by lot system.

Answer both subdivisions (a) and (b)

1. (a) A man is paid at the hourly rate of Rs. 15 per hour for the first 50 hours worked. Thereafter, overtime is paid at 1.5 times the hourly rate for the next 25 hours and 2 times the hourly rate for further hours worked per week. Calculate and print his gross weekly wage.

   (b) Write a C program to implement PUSH and POP operations on stack.

---------- Cut here ----------

2. (a) Write a C program to sort 10 numbers in ascending order.

   (b) Write a C program for linked list implementation of queue operations.

3. (a) Write a C program to compute commission earned by a salesman according to the scheme given below :

Sales Amount	Commission
Upto 10,000	5%
Above 10,000 to 50,000	10%
Above 50,000	15%

(b) Write a C program to evaluate the given mathematical expression using stack.

---------- Cut here ----------

4. (a) Write a C program to find the biggest of 5 numbers. Modify the program to find the biggest of 10 numbers.

(b) Write a C program to implement insert and delete operations an linked list.

---------- Cut here ----------

5. (a) Write a C program to convert a given number into words for numbers 1 to 5. Ex : 1 to ONE, 2 to TWO ....

(b) Write a C program to implement insert and delete operations on queue.

---------- Cut here ----------

6. (a) Write a C program to concatenate two given strings and find the length of the concatenated string.

(b) Write a C program to sort the given set of elements using insertion sort.

7. (a) Write a C program to find the factorial of a given number using function.

(b) Write a C program to sort the given set of elements using selection sort.

---------- Cut here ----------

8. (a) Write a C program to find the product of two matrices.

(b) Write a menu driven program in C to find an element using linear and binary search.

DE–9032

Sub. Code
26

DISTANCE EDUCATION

B.Sc. (CS)/ B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

LAB II – C++ PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by LOT system.
Each sub-division carries 50 marks.

1. (a) Write a temperature conversion program that gives the user the option of converting Fahrenheit to Celsius or Celsius to Fahrenheit. Then carry out the conversion. Use floating point numbers.

   (b) Create a class Employee that contains an Employee number, Employee name and address. Write a Menu driven C++ program to get the 'n' number of employee details and display all details in employee name-wise sorted order.

---------- Cut here ----------

2. (a) Using operator overloading, write a C++ program to find the different and total length of given two various tubes specified in meters and centimetres.

   (b) Create an equivalent of a four-function calculator. The program should request the user to enter a number, an operator, and another number. It should then carry out the specified arithmetical operations: adding, multiplying, subtracting, dividing the two numbers. Finally it should display the result. When it finishes the calculation, the program should ask if the user wants to do another calculation.

3. (a) Using Pointers, create a class and write a program to get the n names and display them in sorted order.

(b) Create a class Int to overload all five integer arithmetic operators (+, -, *, / and %) so that they operate on objects of type Int. If the result of any such arithmetic operation exceeds the normal range of int's - from -32,768 to 32,767 - have the operator print a warning and terminate the program. Write a program to test this class.

---------- Cut here ----------

4. (a) Raising a number n to power p is the same as multiplying by itself t times. Write a function called power( ) that takes a double value for n and an int value for p, and returns the result as double value. Use default argument of 2 for p, so that if this argument is omitted, the number will be squared. Write a main( ) function that gets values from the user to test this function.

(b) Write a program using friend function frifunc( ) which can act on the classes alpha and beta. Using constructors fix the values for alpha and beta.

---------- Cut here ----------

5. (a) Write a program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

(b) Create a C++ class for a stock item abstract data type. It should have the attributes of stock levels (an integer) and unit price (a float). Define the methods to t return the values of these two attributes and to set them using parameters. Add two more methods to allow stock receipts and issues updating the stock level as appropriate. Write a menu driven C++ program to solve the problem.

6. (a) Write a program to process students marks with the help of classes. The class has private variables, for name, mark1, mark2, mark3. It has two member functions – getdata( ) - to get input, - result( ) - to print the results. All subjects mark must be >=50 for pass otherwise fail.

(b) Write a program that reads a group of numbers from the user and places them in array of type float. Once the numbers are sorted in the array, the program should average them and print the result. Use pointer notation whenever possible.

DE–9033

Sub. Code
27

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

COMPUTER LAB – III — APPLICATION PROGRAMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select TWO questions one from Section A and another from Section B.

The questions are given to the candidates by lot system.

SECTION A — (50 marks)

1. Prepare a news report using two columns, insert a picture in the first column and make the text flow around it.

---------- Cut here ----------

2. Type lecture notes and provide audio explanation with the help of sound files.

---------- Cut here ----------

3. Create a worksheet with the following columns:

   Salesman number, Salesman Name, City, Product name, Sale amount.

   Add three records for five different sales man who have carried out sales of different products in different cities. Find the following:

   (a) Citywise total sales

   (b) Salesmanwise total sales

   (c) Productwise total sales

   (d) Grand total sales

SECTION B — (50 marks)

MS–ACCESS

4. Create Inventory table with fields Itemno, Name, Quantity on Hand, Reorder Level.

   (a) Create a form in custom format

   (b) Create a query Reorder to show all the items

   (c) Create a macro that executes Reorder query automatically.

---------- Cut here ----------

5. Create employee table with the fields Empno, Name, Age, Sex, Street, City, Pin, Salary.

   (a) Add data for 20 employees

   (b) Write a query to display all the male employees whose salary is between 3000 and 8000 and living in city "Madurai"

   (c) Write a query to display all the female employees whose ages are in the range 50-60

   (d) Show all the records in the table for the city "Trichy" by filtering

---------- Cut here ----------

6. Create STUDENT table with the following fields: Regno, Name, Mark 1, Mark 2, Mark 3.

   Create ADDRESS table with fields Regno, Street, City and Pin.

   (a) Write a query to display Regno, Name and total of all the three subject marks.

   (b) Write a query to display Regno, Name, Street, City, Pin and total of all the three subject marks.

DE–9034

Sub. Code
36

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

LAB IV – VISUAL BASIC AND ORACLE

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select TWO questions. ONE from Visual Basic and another from Oracle. The questions are given to the candidates by lot system

VISUAL BASIC

1. Write a Visual Basic program to count the number of positive numbers and negative numbers in a given set of numbers

---------- Cut here ----------

2. Write a Visual Basic program to check whether the string is Palindrome or not.

---------- Cut here ----------

3. Write a Visual Basic program to find the largest number and smallest number in a given numbers.

---------- Cut here ----------

4. Write a Visual Basic program that allows the user choice among four arithmetic operations of addition, subtraction, multiplication and division with two given numbers.

5. Write a Visual Basic program to compute the surface area and volume of a sphere given the radius using option button and INPUT boxes.

[Formula: $A = 4 * 3 * R\hat{}2, V = 4/3 * 3{,}14R\hat{}3$]

---------- Cut here ----------

6. Write a Visual Basic program to print twenty address labels in a two-column format for the address.

THE REGISTRAR

ALAGAPPA UNIVERSITY

KARAIKUDI – 63003.

---------- Cut here ----------

7. Write a Visual Basic program to perform Simple Interest and Compound Interest using Command buttons

---------- Cut here ----------

8. Write a Visual Basic program to arrange the numbers in Descending order.

---------- Cut here ----------

9. Write a Visual Basic program to count the number of vowels (a, e, i, o, u) in a string.

10. Write a Visual Basic Program to compute and print either the SUM or the PRODUCT of the first N natural numbers using option buttons.

---------- Cut here ----------

ORACLE

11. Create table ADDRESS with the following Structure:

Field Name	Type	Width
NAME	Character	20
ADDRESS - I	Character	20
ADDRESS — II	Character	20
CITY	Character	20
DISTRICT	Character	20
PIN	Numeric	6

(a) Add 10 Records.

(b) Display all the records in the Table.

(c) Add the field PHONE_NUMBER.

(d) Display all the names that live in CHENNAI.

(e) Show data in fields NAME, CITY and PHONE-NUMBER alone on the screen.

---------- Cut here ----------

12. Create table BIODATA with the following Structure:

Field Name	Type	Width
NAME	Character	25
AGE	Numeric	2
SEX	Character	1

DEGREE	Character	8
CITY	Character	20
PHONE_NUMBER	Numeric	10

(a) Add 5 Records, (BIODATA of 5 Employees).

(b) Display all records of FEMALE Employees.

(c) Delete all B.E. Degree holders and then display the contents.

(d) Remove all Employee records whose AGE is greater than 30.

(e) Display all records having AGE < 25 and CITY is CHENNAI.

13. Create table PAY with the following Structure:

Field Name	Type	Width	Decimal
EMP_NO	Numeric	8	
NAME	Character	25	
DESIGNATION	Character	10	
DOJ	Date		
BPAY	Numeric	8	2
DA	Numeric	8	2
HRA	Numeric	8	
DEDU	Numeric	8	
GPAY	Numeric	8	2
NPAY	Numeric	8	2

(a) Add 5 Records, (Fill all fields except DA, HRA, DEDU, GPAY and NPAY).

(b) Fill DA with 75% of BPAY for all employees.

(c) Fill HRA with 300 for all employees.

(d) Fill DEDU with 750 for all employees.

(e) Fill GPAY with the sum of BPAY, DA and HRA.

(f) Fill NPAY with GPAY-DEDU.

(g) Display all records having salary greater than 30,000.

---------- Cut here ----------

14. Create table ADDRESS with the following Structure:

Field Name	Type	Width
NAME	Character	20
ADDRESS	Character	20
ADDRESS - II	Character	20
CITY	Character	20
PIN	Numeric	6
PHONE NUMBER	Numeric	10

(a) Add 5 Records.

(b) Display all the records in the Table.

(c) Add the field EMAIL_ID.

(d) Display the names whose starting letter is "A".

(e) Change the PIN to 630003 where city is CHENNAI and PIN is 600047.

---------- Cut here ----------

15. Create table LABEL with the following Structure:

Field Name	Type	Width
NAME	Character	20
STREET	Character	20
CITY	Character	20
PIN	Numeric	6

(a) Add 10 Records.

(b) Display the structure of the table.

(c) Display the Descending order sorted list with NAME as primary key.

(d) Display the Ascending order sorted list with CITY as primary key.

(e) Display the contents with appropriate HEADINGS

---------- Cut here ----------

16. Create table PERSONAL with the following Structure:

Field Name	Type	Width
SNO	Numeric	8
NAME	Character	20
CITY	Character	20

(a) Add 10

PIN	Numeric	6
PHONE	Numeric	10

Records.

(b) Display all Names.

(c) Display all names in uppercase, lowercase and initial capital letters.

(d) Display the contents of the table.

(e) Display all names and length of names.

17. Create table SALES with the following Structure:

Field Name	Type	Width	Decimal
SALE_NO	Numeric	5	
SALE_NAME	Character	20	
SALE_AMT	Numeric	8	2
COMM	Numeric	6	2
SALE_DATE	Date		

(a) Add 5 Records.

(b) Replace COMM with the following conditions:

If Sales Amount is less than 1000 there is no commission.

If Sales Amount is greater than or equal to 1000 but less than or equal to 5000, commission is 5% of sales amount.

If Sales Amount is greater than 5,000, commission is 10%.

(c) Display all the records.

(d) Display the records having commission < 1000.

(e) Display the records having commission> 1000.

(f) Display the Average sales and Average Commission.

---------- Cut here ----------

18. Create table BLOOD with the following Structure

Field Name	Type	Width
DNO	Numeric	8
NAME	Character	20
DOB	Date	
DSEX	Character	1

(a) Add 10

DBLOOD	Character	3
DADDR1	Character	20
DADDR2	Character	20
DCITY	Character	20
DPIN	Numeric	6
DPHONE	Numeric	10

Records.

(b) Display all the records.

(c) Display the information of donors in the age group 20 to 25 using BETWEEN options.

(d) Display the information of donors having date of birth BETWEEN 01-JAN-2001 and 31-DEC-2013.

(e) Display the information of female donors with age between 20 and 25 using BETWEEN options.

---------- Cut here ----------

19. Create table BLOOD with the following Structure:

Field Name	Type	Width	Decimal
NUM	Numeric	6	2
ABSNUM	Numeric	6	2
SQRTNUM	Numeric	6	2
ROUNDNUM	Numeric	6	2
TRUNCNUM	Numeric	6	2
SQRNUM	Numeric	6	2
CUBENUM	Numeric	6	2

(a) Add S Records, (Some numbers must be Negative or Decimal)

(b) Display all the records.

(c) Replace ABSNUM with absolute value of given numbers.

(d) Replace ROUNDNUM by using ROUND function.

(e) Replace TRUNCNUM by using TRUNC function.

(f) Replace SQRNUM with square of given numbers.

(g) Replace CUBENUM with cube of given numbers.

---------- Cut here ----------

20. Create table MARK with the following Structure:

Field Name	Type	Width
REQ NO	Numeric	8
NAME	Character	20
TAMIL	Numeric	3
ENGLISH	Numeric	3
MATHS	Numeric	3
SCIENCE	Numeric	3
SOCIAL SCIENCE	Numeric	3
TOTAL	Numeric	3

(a) Add 10 Records.

(b) Show data in fields REG No, Name and TOTAL alone on the screen.

(c) Show data in fields REG No, Tamil and ENGLISH alone on the screen.

(d) Show data in fields REG No and TOTAL alone on the screen.

(e) Show all fields on the screen.

DE–9035

Sub. Code
37

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, DECEMBER 2015.

LAB V – JAVA PROGRAMMING

(2007 Onwards)

Time : Three hours Maximum : 100 marks

One question should be selected by the examiner for each candidate by Lot system.

Each sub-division carries 50 marks.

1. (a) Write a Java program to create multiple threads.

   (b) Write a Java program to read and write the data using random access file.

---------- Cut here ----------

2. (a) Write a Java program to create packages.

   (b) Write a Java program to display the contents of file along with a line number.

---------- Cut here ----------

3. (a) Write a Java program to multiply two given matrices.

   (b) Write a Java program that displays the number of characters, lines and words in a text file.

---------- Cut here ----------

4. (a) Write a Java program to arrange the names in alphabetical order using command line arguments.

   (b) Write a Java program that reads a file name from the user and displays the contents of file.

5. (a) Write a Java program to illustrate over loading and overriding methods.

   (b) Write a Java program to draw the various polygons using applet.

---------- Cut here ----------

6. (a) Write a Java program to implement the single and multi level inheritance.

   (b) Write a Java program to count the number of words and characters in a text.